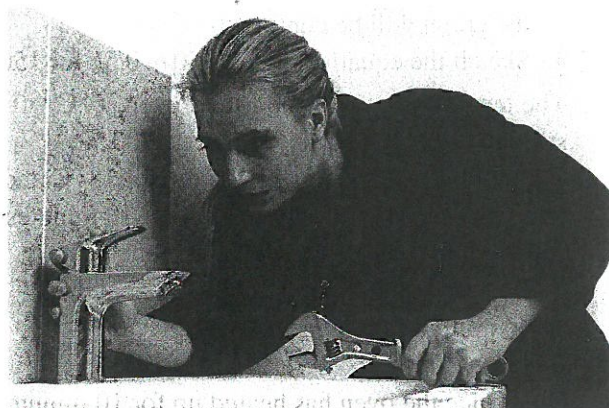


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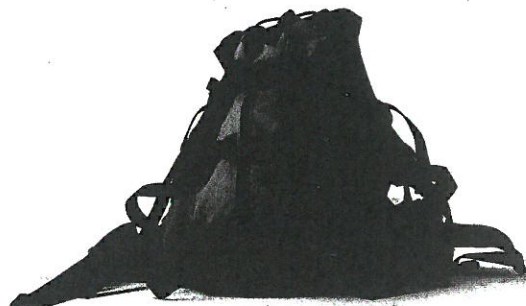
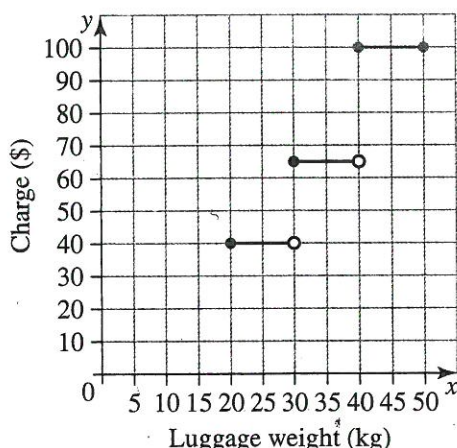
7. The following table shows the costs to hire a plumber.

Time (minutes)	Cost (\$)
0–15	45
15–30	60
30–45	80
45–60	110

- Represent this information on a step graph.
- Anton hired the plumber for a job that took 23 minutes. How much will Anton be expected to be charged for this job?



8. Airline passengers are charged an excess for any luggage that weighs 20 kg or over. The following graph shows these charges for luggage weighing over 20 kg.



- How much excess would a passenger be charged for luggage that weighs 31 kg?
 - Nerada checks in her luggage and is charged \$40. What is the maximum excess luggage she could have without having to pay any more?
 - Hilda and Hanz have two pieces of luggage between them. One piece weighs 32 kg and the other piece weighs 25 kg. Explain how they could minimise their excess luggage charges.
9. **WE15** The following two equations represent water being added to a water tank over 15 hours, where w is the water in litres and t is the time in hours.

$$\text{Equation 1: } w = 25t, 0 \leq t \leq 5$$

$$\text{Equation 2: } w = 30t - 25, 5 \leq t \leq 15$$

- Determine how many litres of water are in the tank after 5 hours.
 - At what rate is the water being added to the tank after 5 hours?
 - For how long is the water added to the tank at this rate?
 - Sketch a piecewise graph to represent the water in the tank at any time, t , over the 15-hour period.
10. A car hire company charges a flat rate of \$50 plus 75 cents per kilometre up to and including 150 kilometres. An equation to represent this cost, C , in dollars is given as $C = 50 + ak, 0 \leq k \leq b$, where k is the distance travelled in kilometres.
- Write the values of a and b .
 - ~~Using CAS, a spreadsheet or otherwise~~, sketch this equation on a set of axes, using appropriate values. The cost charged for distances over 150 kilometres is given by the equation $C = 87.50 + 0.5k$.
 - Determine the charge in cents per kilometre for distances over 150 kilometres.
 - By solving the two equations simultaneously, find the point of intersection and hence show that the graph will be continuous.
 - Sketch the equation $C = 87.50 + 0.5k$ for $150 \leq k \leq 300$ on the same set of axes as part b.